

Infrastructure / Offline



Trusted Authority (TA)

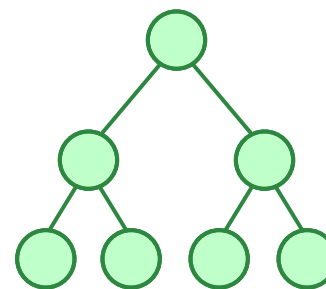
Vehicle Registration,
Revocation Decisions

Revocation Commands,
Policy Updates



AST Issuing Service (AIS)

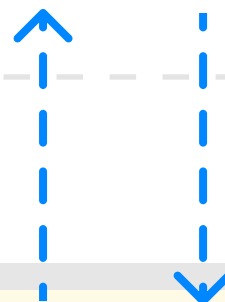
Issues ASTs,
Maintains Merkle Tree



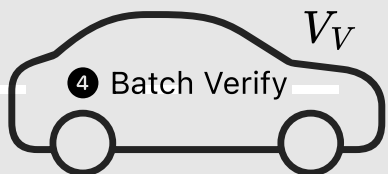
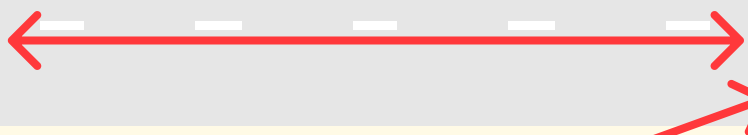
V2V / Online

❶ AST Request
(via RSU/Cellular)

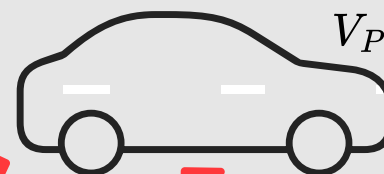
❷ AST + Merkle Path



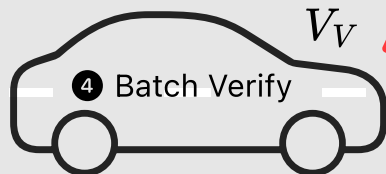
❸ BSM + ZKP Proof π + Merkle Root R



❹ Batch Verify

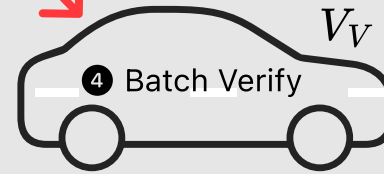


V_P



❹ Batch Verify

V_V



❹ Batch Verify

V_V

